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Exploiting auditory filter models as interpretable convolutional frontends to obtain optimal architectures for speaker gender recognition

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ABSTRACT

Identifying the gender of a speaker is an important problem which can be considered as a preliminary task in many speech processing applications such as speaker identification and emotion recognition. Although many accurate models have been developed through the progress of Deep Neural networks (DNNs), the computational complexity of these models makes them unsuitable for some situations with limited hardware resources. Pruning the designed architecture of DNNs to achieve the same initial accuracies by reducing the number of network parameters is a way to create more efficient architectures. In this paper, Auditory Filter Models (AFMs) are exploited to be used as frontend convolutional filters. In addition to the fact that these meaningful filters can be represented by a few parameters, they also provide better performance compared to traditional conventional filters. The proposed method can increase efficiency, improve the generalizability of the models, and reduce the number of parameters for gender recognition. Moreover, by utilizing a clustering approach to reduce the number of learnable filters of the AFM in the first convolutional layer, the network parameters can be reduced even further. Experimental results conducted on four well-known datasets, namely *TIMIT*, *LibriSpeech*, *VoxCeleb1,* and *RAVDESS*, prove the validity of the claims made earlier. The results show that the use of this technique not only enhances the performance of the models but also facilitates the learning of a rich and informative representation of speech data. This finding emphasizes the effectiveness and robustness of the proposed approach in handling diverse datasets and real-world scenarios.

1. Introduction

Speaker Gender Recognition (SGR) is a helpful primary task in some applications such as human–machine interaction, speech recognition, crime detection, and emotion recognition [1,2]. Moreover, it can be used as a pre-processing task in some speech-processing applications, like speaker identification, to achieve better performance [3]. In such cases, early diagnosis of gender speeds up the overall process.

Fortunately, with the emergence of the Deep Learning (DL) field of research, many effective methods have been developed for this purpose. However, DNNs typically have billions of parameters, which leads to significant computation complexity, memory usage, and power consumption. Deploying such large DNNs on devices with limited hardware resources is challenging [4]. Designing an optimal architecture with almost the same accuracy as the initial (primary) model is a solution to overcome this problem.

Interpretability is the ability to explain or present the model's behavior in understandable terms for humans [5]. Many DNNs have an important problem called un-interpretability. In other words, the internal functionality of deep architectures is ambiguous, and the model has the nature of a black box [6]. This property makes DNNs unreliable, especially in forensic and medical applications. Two distinct approaches have been discussed in the literature to achieve model understanding. The first approach involves employing inherently interpretable models, such as Decision Trees; however, these models often lack sufficient accuracy. Indeed, there exists a trade-off between accuracy and interpretability when constructing models that are designed to be easily understood. Simpler models tend to offer higher interpretability, while as the complexity of the model increases, its interpretability diminishes. In cases where the necessity arises to utilize a complex black-box model, the second approach, which involves providing explanations for pre-built models in a post hoc manner, can be quite advantageous [7]. In this fashion, the explanations serve as descriptive insights into the model's behavior, offering humans a clearer perspective on its functioning. Using Auditory Filter Models (AFMs) in the first layer of a Convolutional Neural Network (CNN) is a way to create interpretable

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CNNs for speech-based applications [8–10]. In this way, the model's sensitivity to different frequency bands in the desired task can be understood in a post hoc explanation manner, and the number of learnable network parameters is reduced [10].

The input of DNN models used for gender recognition can be categorized into two main approaches. One approach uses numeric features of the speech signal, such as mean, standard deviation, and mode frequency, while the other benefits from spectral features like *Mel*-Frequency Cepstral Coefficients (MFCC) and *Log-Mel* features [11]. This paper selects a general CNN architecture for use in the SGR task. Unlike traditional approaches that use numeric or spectral features as input, this model's input is the raw speech signal. This property helps the model automatically extract important features from the raw signal. To achieve an efficient model, the filters learned in the first convolutional layer of the architecture are restricted to be from the most common AFMs. By emulating each learned filter by a pole location in the complex plane, good insight into the frequency distribution and bandwidth of the learned filter bank is obtained. Since some learned filters operate in the same frequency range, a clustering algorithm is then used to group similar filters into a more general cluster. In the next step, the resulting reduced cluster centers are selected as the initial filters of a new model. The new interpretable learned model has fewer parameters with almost the same accuracy.

The proposed model shows several distinctive features that distinguish it from the existing models. Some key differentiating factors are: (1) *Simplicity*: It employs a typical end-to-end convolutional architecture with raw speech signal as its input. (2) *Interpretability*: The inclusion of AFMs enhances the model's interpretability, in a post hoc explanation manner, enabling it to focus on important regions or features in the data. This mechanism facilitates a better understanding of the learned representations and aids in capturing essential characteristics of the input data. Furthermore, it reveals the frequency analysis performed by the human auditory system for the SGR task. (3) *Generalizability*: The model applies specific constraints, particularly on the first convolutional layer, which encourages learning more generalized representations. It can handle diverse datasets and real-world scenarios effectively, allowing for reliable performance. (4) *Enhanced representation learning*: The utilization of AFMs leads to the achievement of richer and more informative representations within the convolutional layers. This leads to improved feature extraction and subsequent task performance.

The remainder of the paper is organized as follows: Section 2 provides an overview of related works, including traditional approaches and DNNs for the SGR task and interpretable DNNs for speech processing. The proposed method based on the AFMs is described in detail in Section 3. Experimental results and discussions are presented in Section 4. Finally, Section 5 concludes the paper and suggests directions for future research.

2. Related works

Several standard approaches have been employed for the SGR task. One common method involves extracting traditional features such as fundamental frequency and cepstral features from the speech signal, followed by training a binary classifier. For instance, in the works of [12,13], the authors extracted pitch, energy, and MFCC features from speech signals. Then these features were used to classify the samples using a Support Vector Machine (SVM) classifier [14]. In another study conducted by [15], a combination of features, including fundamental frequency, spectral entropy, spectral flatness, mode frequency, and MFCC features was utilized. The authors employed SVM and K-Nearest Neighbor (KNN) classifiers to classify the samples based on these features. Linear Predictive Residual (LPR) and MFCC features with a classifier based on Gaussian Mixture Model (GMM) were presented in [16]. In [17], a combination of acoustic features and MFCCs was employed as the feature set, using an SVM classifier. The statistical features of the pitch were exploited in [18] by examining some classical classification methods like CNN, SVM, Multi-Layer Perceptron (MLP), and Logistic regression (LR).

In the following, we will discuss recent methods based on DL approaches for SGR. In [19], a gender detection system was proposed for describing gender equality in audio-visual streams. The system used MFCC features as input and utilized different models for speech representation, such as GMMs, i-vectors, and CNNs. A simple CNN architecture for gender recognition based on the raw speech signal was proposed by [20]. Reference [21] used a double-layer Long Short-Term Memory (LSTM) structure for gender recognition. The input features used in the study were numeric features selected based on the relief feature selection method [22].

Reference [23] introduced a classification model for language-independent SGR, based on *ResNet-50* [24], which utilized spectrograms as input features. In the study conducted by [25], a model based on transfer learning was proposed for identifying gender in movie audio. This paper used subtitle text to detect segments with voice activity using an RNN architecture accurately. The feature embeddings obtained from a pre-trained audio classification model were then utilized for gender prediction.

In the research conducted by [26], ensemble learning techniques were combined with a semi-supervised learning framework to develop more stable and robust gender classifiers. The age information of the speaker was used as auxiliary information in [27] for improving gender identification in a multi-task learning framework. [28] evaluated the effectiveness of different DNN-based embedder architectures, such as x-vector and d-vector, for age estimation and SGR tasks. The authors employed a transfer learning approach by pretraining the embedder network for a speaker recognition task and then fine-tuning it to estimate joint age and SGR. In [29], various network topologies were investigated to assess the influence of model architecture on the performance of joint age and SGR tasks.

A gender-dependent emotion recognition system was proposed in [30], where the gender recognition module worked based on GMMs. In [1], a multi-head attention DL architecture was proposed for both speech emotion and gender recognition based on log *Mel*-filter bank energy spectral features. A similar approach with a different architecture was utilized in [31], which used speech spectrogram as input features to represent the emotions better. In both of these works, the gender recognition task was applied as an auxiliary task that helps to learn the gender-specific features that influence the emotional characteristics in speech, resulting in improved accuracy of speech emotion recognition.

The effectiveness of self-attention models in dealing with voice gender recognition in unconstrained environments has been studied in [11]. This paper used MFCC as input features and evaluated its proposed models in environments such as background noise and different languages, accents, ages, and emotional states of the speakers. Reference [32] presents an end-to-end CNN for age and gender recognition with a multi-attention module that helps to extract spatial and temporal salient features from the input data. The input of the model was the spectrogram generated from the speech signal, and rectangular filters were used in convolutional layers. The spectrogram was also used in [33] as an input representation of the speech signal to a CNN model for accent and gender recognition tasks. A hybrid SGR system that used both human-generated features and a deep model extracted features was presented in [34]. The deep model used in this research was a combination of CNNs and Gated Recurrent Units (GRU) to extract locally encoded important features and resolve vanishing and exploding gradient problems at the same time.

The research presented by [35] proposed a speech signal representation in the time–frequency domain using Linear Chirplet Transform (LCT). This representation yielded a feature vector that was sensitive to frequency changes in human speech. The authors utilized the *LeNet* architecture [36] for classification. They demonstrated that features based on LCT outperformed those based on MFCC and Fourier transform in the SGR task.

In recent studies, it has been demonstrated that incorporating meaningful filters in the first layer of a CNN architecture for speech processing tasks can improve model interpretability, reduce the number of network parameters, and increase model accuracy. This technique was introduced in the *SincNet* paper [8], which used rectangular filters in the first layer for speaker identification and verification tasks. In [9], the authors proposed the use of three additional filter types, including triangular, Gaussian, and gammatone filters to further improve the performance. The resulting networks were referred to as *Sinc2Net*, *GaussNet*, and *GammaNet*, respectively. In a more recent study, presented in [10], a type of *"Gaussian cascaded filter"* was added to the aforementioned filter types, and the performance of the filters was analyzed based on its similarity to the filtering mechanism of the human auditory system.

Making compact DNNs is also an important issue that arises in some low-resource applications. In such situations, networks must be optimized for an accuracy-speed trade-off. Network pruning is one approach to designing efficient neural networks by removing unimportant weights, filters, channels, or layers from the primary architecture. A survey on efficient CNNs and hardware acceleration methods, focusing on quantized models, optimized architectures, and resource-constrained systems, was presented in [4].

3. Proposed method

In this section, we will explain the details of the proposed method. First, we will introduce the model architecture and then discuss the techniques for obtaining an optimal architecture.

3.1. Model architecture

The proposed method employs the same primary architecture as in [8], which is illustrated in Fig. 1. Prior to feeding the speech signals into the model, the silent regions are removed in a pre-processing step. The input to the model consists of 200 ms speech chunks with a temporal shift of 10 ms. Then, three consecutive convolutional layers with Leaky-ReLU activation functions [37] are used. After each convolutional layer, a pooling layer with length 3 and a layer normalization [38] are applied. The learned features from the convolutional layers are flattened and passed to the last three fully-connected layers, each of which is followed by a batch normalization [39] layer. Finally, a SoftMax layer is used to obtain the class probability.

3.2. Network pruning

While DNN models achieve high performance in gender recognition tasks, the computational resources they require make them impractical for low-resource applications. To address this issue, there are methods for designing efficient CNN models. In this study, we aim to modify the primary model to obtain a lighter model with the same accuracy. This is achieved through two main steps: using AFMs in the first convolutional layer and selecting the best learned filters in the first layer. These steps are described in the following subsections.

3.2.1. Using auditory filter models (AFMs)

Most of the advances in artificial intelligence have been achieved by mimicking the functionality of various parts of living beings. In this section, we will begin by elucidating the auditory mechanism in humans. Subsequently, we will illustrate how this intricate mechanism can be effectively translated into a straightforward CNN architecture.

The peripheral part of the auditory system of most mammals is similar. This part includes the outer ear, middle ear, and inner ear. The outer ear significantly alters the incoming sounds, especially high frequencies. Sound waves reach the eardrum through the auditory canal and cause it to vibrate. These vibrations reach the cochlea through the middle ear by three small bones (ossicles). The cochlea is covered with two membranes called Reissner's and Basilar. The response of the basilar membrane (BM) to a sinusoidal stimulus is in the form of a traveling wave that goes from the beginning to the end of the BM. The response of BM to sounds with different frequencies is different at the beginning and end of the cochlea. At the beginning, named the base, the responses are stiff and narrow, and at the end, called the apex, they become much less stiff and wider. As a result, the location of the peak in the vibration pattern varies according to the excitation frequency. Functionally, the cochlea acts like a Fourier analyzer with a more limited frequency analysis capability. In other words, it can separate the sinusoidal components of the complex signal from each other. The frequency that produces the greatest response at a point on the BM is called the characteristic frequency (CF) for that location. Each point on the BM can be considered a band-pass filter with a different center frequency and bandwidth. The BM response in different parts does not have the same bandwidth; But the larger the CF, the greater the bandwidth. Various experiments have shown that the human auditory system acts as if it consisted of a bank of overlapping band-pass filters. These filters are called audio filters. Each location in the BM responds to a limited range of frequencies, so each point corresponds to a filter with a different

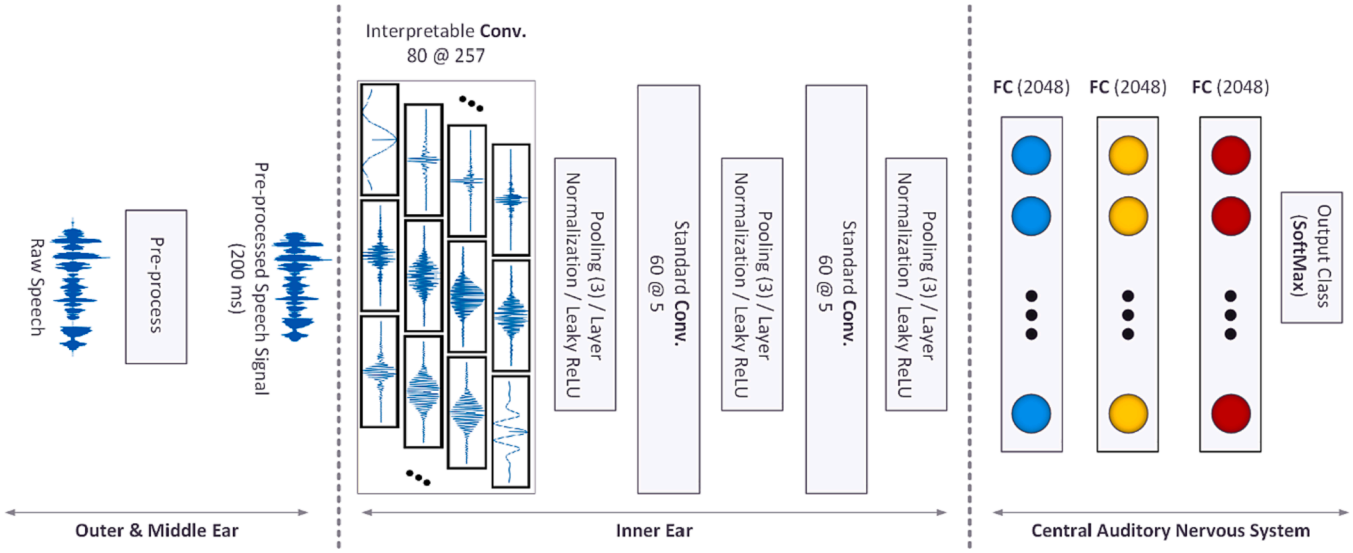


Fig. 1. CNN model architecture [10].

frequency center [40].

The overall architecture of typical CNNs can be matched to the human auditory system. The first part of the model architecture, which applies preprocessing steps, corresponds to the filtering mechanism of the outer and middle ear in the human auditory system. The spectral analysis of the cochlea in the inner ear is mimicked by the convolutional layers. Finally, the DNN layers correspond to the central auditory nervous system in the brain.

Psychologists and physiologists have proposed various approaches for models of auditory filtering [41]. Various types of filters have been introduced in [41] for describing auditory function, including simple resonances, Gaussian filters, rectangular filters, roex family, gammatone family, and cascade filter family.

It has been demonstrated in [10] that incorporating some of the filter types mentioned above in the first convolutional layer can significantly enhance the performance of the model in speaker identification tasks in terms of accuracy and number of parameters. Based on the arguments presented in that work, *rectangular*, *triangular*, *Gaussian*, *gammatone*, and *cascade* filters are utilized to be examined here. The formulas of these filter types are presented in Table 1. Since the *Gaussian* model gives the best frame-level accuracy, we use a cascade of two *Gaussian* filters in the *Cascade* model. In these equations, A represents the amplitude, B is the bandwidth of the filter, and f_c is the center frequency. In all of them, the amplitude and two cutoff frequencies f_1 and f_2 are learned for each filter which $B = f_2 - f_1$ and $f_c = (f_1 + f_2) / 2$. As expected, the number of parameters is doubled in the case of cascade filters.

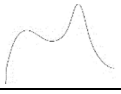
The number of parameters that need to be learned in a typical convolutional layer with N filters and a filter length of L is $L \times N$. However, in the proposed model with meaningful AFMs in the first convolutional layer, only $3 \times N$ parameters need to be learned. This means that the number of learning parameters is reduced by a factor of $3 / L$.

3.2.2. Filter pruning

As previously mentioned, each meaningful learned filter can be represented by only three parameters: Amplitude (A), Bandwidth (B), and center frequency (f_c). On the other hand, each band-pass filter can be viewed as a rational transfer function in the z-domain as in Equation (1). Since we are dealing with a real-valued system, two poles that are complex conjugates of each other are considered here, a and a^*. Two zeros are located at the origin ($Z = 0$). The poles can be mapped to the complex plane (z-plane) based on its bandwidth and center frequency using Equation (2) as follows,

$$H(Z) = \frac{Z^2}{(Z-a)(Z-a^*)} = \frac{1}{(1-aZ^{-1})(1-a^*Z^{-1})} \tag{1}$$

Table 1
Details of the used Auditory Filter Models (AFMs).

Filter type	Time domain equation	Frequency response shape
Rectangular Filter (*Sinc*)	$h_{rect}[n] = 2AB\text{sinc}(Bn)\cos(2\pi f_c n)$	
Triangular Filter (*Sinc2*)	$h_{tri}[n] = A\text{sinc}^2(Bn)\cos(2\pi f_c n)$	
Gammatone Filter (*Gamma*)	$h_{gamma}[n] = An^{N-1}\exp(-2\pi Bn)\cos(2\pi f_c n)$	
Gaussian Filter (*Gauss*)	$h_{gauss}[n] = A\exp(-n^2/2\sigma^2)\cos(2\pi f_c n)$	
Gaussian Cascade Filter (*Cascade*)	$h_{cascade}[n] = h_{gauss,1}[\frac{n}{2}] * h_{gauss,2}[\frac{n}{2}]$	

$$a = e^{-B+jf_c} \tag{2}$$

where the imaginary unit is denoted by j. This mapping is demonstrated in Fig. 2 for three different sample gammatone filters. The two vertical lines within the figure serve to visually represent the cutoff frequencies that are acquired through the training process.

The mapping process can be extended to all learned filters, enabling the visualization of pole distributions across the complex plane within the learned filter bank. The resulting representation accurately reflects the frequency behavior of the filters at all frequencies (as shown in Fig. 3a). In this new representation of the learned filter bank, some poles are located close to each other. The corresponding filters of these poles have a similar impact on the learned representation in the convolutional layers of the CNN model. Therefore, it is possible to ignore these redundant filters by combining nearly located poles into a single new pole without losing any essential information (as shown in Fig. 3b). This process can be facilitated by employing a clustering algorithm. The result of this process is to obtain a new filter bank with a lower number of band-pass filters, which is called a reduced filter bank here.

After evaluating different clustering algorithms, we select the mean-shift clustering approach [42] as a method to eliminate redundant filters. Mean-shift works based on the concept of kernel density estimation. It makes no assumptions about the shape of the distribution or the number of clusters. Many types of kernels can be used in the clustering procedure, but the Gaussian kernel is the most popular one. The kernel bandwidth parameter determines the number of clusters.

We consider the real ($\Re$) and imaginary ($\Im$) parts of the Cartesian representation of each pole as the feature set (D) fed to mean-shift clustering as in Equation (3). In building this feature set, the complex conjugate of the primary pole is not considered.

$$D = \{\Re(a_i), \Im(a_i)\}_{i=1}^{N} \tag{3}$$

where N is the number of initial filters in the filter bank, a_i represents the i'th filter in the filter bank, and $\Re(a_i)$ and $\Im(a_i)$ are calculated by $\frac{a_i + a_i^*}{2}$ and $\frac{a_i - a_i^*}{2j}$, respectively.

The method considers the sample points (pole locations) in the feature space as an empirical density function, where dense regions correspond to local maxima or cluster centers. Local maxima are here obtained through a gradient ascent procedure, and other points are considered members of the clusters based on their distance from the center point.

The resulting cluster centers can be conceptualized as new filters, each characterized by its distinct frequency center and bandwidth. The cluster center, cc, initially represented in Cartesian coordinates, undergoes a transformation to polar coordinates using Equations (4) and (5).

$$f_c = \begin{cases} \tan^{-1}\left(\frac{\Im(cc)}{\Re(cc)}\right) & R(cc) \geq 0 \\ \tan^{-1}\left(\frac{\Im(cc)}{\Re(cc)}\right) + \pi & ,\Re(cc) < 0 \end{cases} \tag{4}$$

$$B = -\ln\sqrt{\Re(cc)^2 + \Im(cc)^2} \tag{5}$$

Through the inverse mapping of the resulted poles back to cut-off frequencies, the reduced filter bank is derived. Since this new reduced filter bank is powerful enough to construct a rich representation of the speech signal, we use it in the first convolutional layer of the CNN model and freeze this layer during the re-training procedure. The new *reduced* model has fewer filters in the first layer, and experimental results show that it achieves a good performance compared to its basic model.

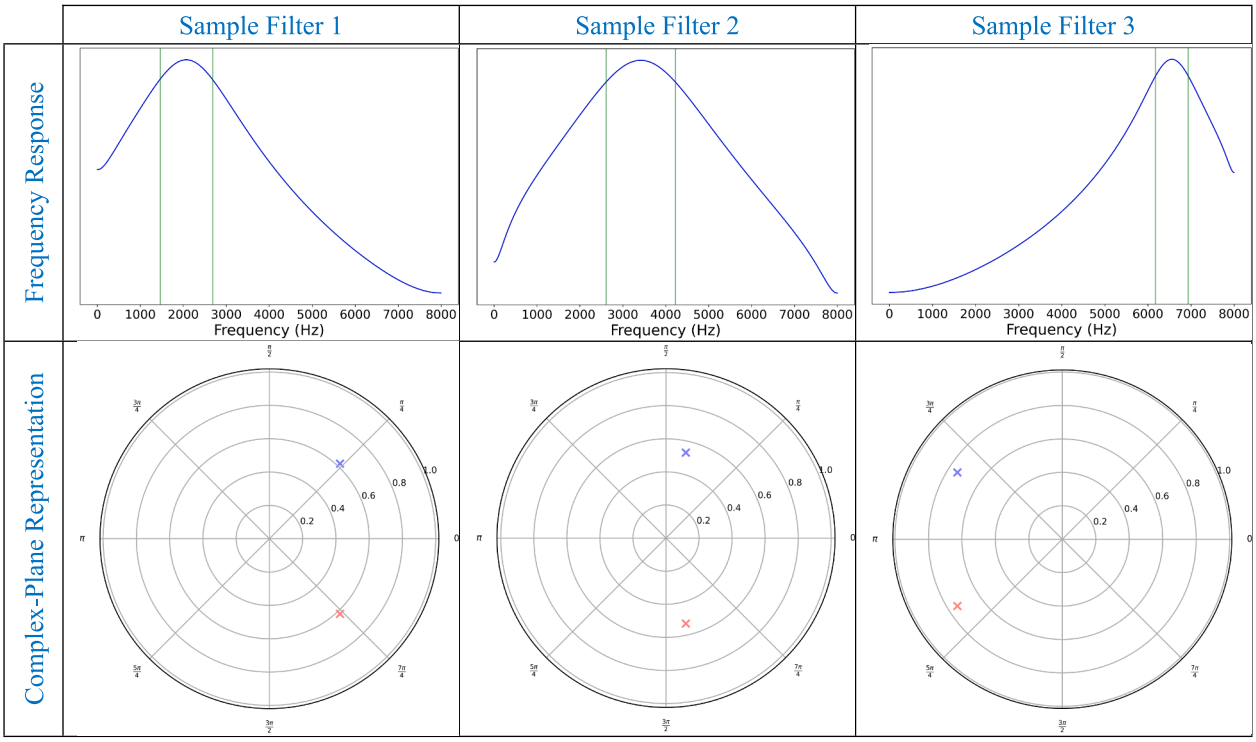
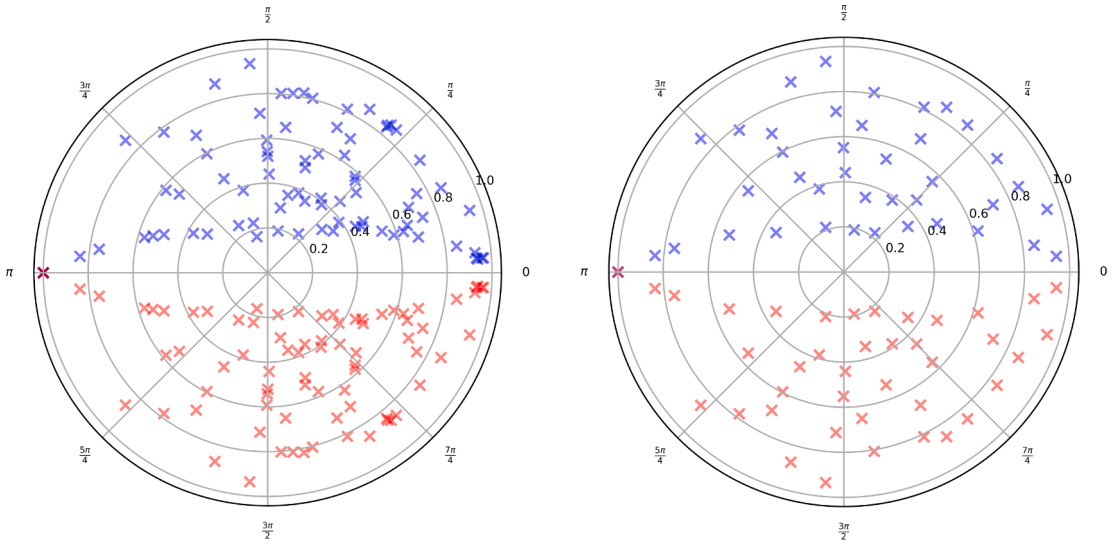


Fig. 2. Three sample cases of mapping of the learned filters to the complex plane.



(a) Initial filter bank (b) Reduced filter bank

Fig. 3. The location of the poles in the complex plane for an initial CNN-based learned filter bank (a) and its reduced filter bank (b).

4. Results and discussions

4.1. Experimental setup

4.1.1. Datasets

This study considers four datasets: the *TIMIT* [43], *LibriSpeech* [44], *VoxCeleb1* [45], and *RAVDESS* [46] datasets. The complete version of the *TIMIT* dataset is utilized for training the models. However, due to the presence of noticeable environmental and stationary background noise in some recordings from the *LibriSpeech* clean corpus, this dataset is employed for fine-tuning the models. Finally, the performance of the models is compared using the *VoxCeleb1* and *RAVDESS* datasets to evaluate their generalizability. The detailed characteristics of each dataset are presented in the subsequent subsections.

4.1.1.1. Timit. The *TIMIT* Acoustic-Phonetic Continuous Speech Corpus, which is a well-known standard speech-based dataset, is used here for SGR. The dataset consists of 16 kHz recordings of 630 speakers from the eight major dialects of American English, with 438 (70%) speakers being male and 192 (30%) being female. This male–female ratio is consistent for both training and testing data. To validate the performance during the training process, 20% of the training set is reserved as validation data. Table 2 provides an overview of the number of samples and speakers in the training, validation, and test sets for the

Table 2
Number of train, validation, and test samples in the *TIMIT* dataset.

TIMIT dataset stat.	Train	Validation	Test	Total
# samples	2960	736	1344	5040
# speakers	370	92	168	630

TIMIT dataset.

4.1.1.2. LibriSpeech. *LibriSpeech* is a dataset consisting of read English speech derived from LibriVox's audiobooks. In this study, we specifically utilize the train-clean-360 subset of the corpus, which contains 16 kHz speech data with approximately equal numbers of male and female speakers. Since the corpus is compiled from audiobooks, there are variations in the tone and style of speech among utterances from the same speaker. Additionally, the presence of background noises in different recordings makes it suitable for fine-tuning the models.

Table 3 provides information on the number of speakers in the train, validation, and test sets for the *LibriSpeech* corpus. To minimize the training data, we randomly select only 12 – 15 s of speech for each speaker. Similarly, the development and test samples consist of 2 – 6 s segments for each speaker.

4.1.1.3. VoxCeleb1. *VoxCeleb1* is a large-scale audio-visual dataset collected from open-source media designed to evaluate the performance of various speech-processing applications in noisy and unconstrained environments. The dataset comprises a substantial collection of 153,516 utterances from 1,251 speakers, of which 690 are male, and 561 are female. In this study, *VoxCeleb1* is specifically chosen to evaluate the performance of the final fine-tuned model, as it ensures that the models learn a representation that can be generalized across diverse scenarios and speaker characteristics.

4.1.1.4. Ravdess. The Ryerson Audio-Visual Database of Emotional Speech and Song (*RAVDESS*) is a comprehensive emotional dataset comprising 1440 utterances performed by 24 professional actors, with an equal distribution of 12 males and 12 females. Each actor conveys 8 different types of emotions, making the dataset highly diverse. The dataset includes three types of files: audio-only, audio-visual, and video-only. For our specific experiments, we focused solely on the audio files. These audio samples have a sampling frequency of 48 kHz. To make them compatible with our model, we resampled all audio files at 16 kHz.

Our primary goal in using the *RAVDESS* dataset is to evaluate the performance of our gender recognition model across various data types and emotions. By working with such a versatile and extensive dataset, we can ensure that our SGR model's capabilities are robust and reliable.

4.1.2. CNN model parameters

In the first convolutional layer of the CNN model (Fig. 1), the model learns 80 filters with a length of 257. The pooled normalized output of the first layer is then fed into the next layer, which has 60 filters with kernels of size 5. The third convolutional layer has the same number of parameters as the second layer. Each of the last fully-connected layers has 2048 neurons. The negative log-likelihood loss and the RMSprop optimizer are here used in the training process for 500 iterations. The mini-batch size is set to 128 and the learning rate is constant and set to 0.001.

Table 3
Number of train, validation, and test speakers in the *LibriSpeech* corpus.

Librispeech corpus stat.	train-clean-360	dev-clean	test-clean
# males	482	20	20
# females	439	20	20
# speakers	921	40	40

4.2. Examining the impact of using the models of auditory filtering

In this section, we report the obtained experimental results and compare the accuracy of the different used AFMs with the baseline methods and literature works. Each method is named based on the filters used in the first convolutional layer.

4.2.1. Performance evaluation

The initial experiments are conducted three times, and the average and standard deviation of the results are reported here. The models are trained on the *TIMIT* dataset with the pre-defined division of train and test data in section 4.1.1.1 and no fine-tuning is done here. The signal level error rate, also known as Classification Error Rate (CER), Frame Error Rate (FER), and corresponding average accuracy (based on the CER) are the evaluation metrics presented in Table 4. In all of the tables presented hereafter, the best and second-best results in each column are highlighted using the bold and underlined formats, respectively. In the table, the *Free-CNN* model refers to the aforementioned CNN architecture, where traditional convolutional filters are learned in the first layer without any constraints. On the other hand, the *Mel-CNN* model utilizes the same architecture, but with the first convolutional layer frozen in the training process and set with 80 *Mel*-scaled filters as its filter bank.

As evident from the results in Table 4, the filter models based on auditory processing methods demonstrate improvements compared to the *free* and *Mel* filter models. The *Mel-CNN* and Cascade methods achieved the worst and best results in both CER and FER metrics, respectively. The use of cascade filters in the first convolutional layer gives an accuracy of over 97% in the frame level, which is a remarkable accuracy for gender recognition based on a 200 ms segment of the speech. Subjective analysis based on listening to misclassified speech signals shows that it is also difficult for humans to distinguish the gender label of the wrong files.

To evaluate the effectiveness of the models proposed in this study, we compare their accuracy with the accuracy reported in the literature for methods applied to the *TIMIT* dataset. There are several basic methods that report their performance on a subset of the TIMIT dataset. These results are presented in Table 5. The input features and algorithm used in each of these baseline methods are presented in the respective columns. In our experiments, we opted for the best-performing model obtained from prior runs. To have a fair comparison, we then proceeded to randomly select 8 males and 8 females from the test set of the TIMIT dataset as its test subset. This process was repeated a total of 10 times for each model and the resulting outcomes were averaged and showcased in Table 5.

Evidently, based on the results of Table 5, the accuracy achieved by our proposed models using the AFMs is better than the baseline methods. Additionally, our proposed models operate in an end-to-end manner and use the unprocessed speech signal as their input. This approach demonstrates that the integration of AFMs into the CNN frontends enhances the distinctiveness of the models. It is noteworthy that this improvement is achieved without the need for any additional feature extraction preprocessing steps.

Table 6 presents the input features, algorithms, end-to-end and

Table 4
CER, FER, and accuracy metrics for auditory filter models on the complete *TIMIT* dataset. The reported results represent the average and standard deviation from three independent runs.

Method	CER (%)	FER (%)	Accuracy (%)
Free-CNN	1.16 ± 0.16	3.14 ± 0.04	98.84
Mel-CNN	1.36 ± 0.09	4.92 ± 0.22	98.64
Sinc	1.16 ± 0.05	2.93 ± 0.34	98.84
Sinc2	0.97 ± 0.08	3.16 ± 0.29	99.03
Gamma	0.99 ± 0.17	2.96 ± 0.07	99.01
Gauss	1.09 ± 0.11	2.90 ± 0.15	98.91
Cascade	**0.84 ± 0.12**	**2.88 ± 0.21**	**99.16**

Table 5
The comparison of the accuracy of auditory filter models and baseline methods on a subset of the *TIMIT* dataset.

Method	Input	Algorithm	end-to-end	Interpretable	Accuracy (%)
[15]	Traditional features	KNN	No	No	96.80
		SVM			87.50
[12]	Pitch + Energy + MFCC	SVM	No	No	96.45
[13]	pitch	SVM	No	No	Below 60
	Signal energy	SVM			Below 60
	MFCC	SVM			69.23
[16]	LPR + MFCC	GMM	No	No	98.00
[18]	Pitch Feature Set	CNN	No	No	99.07
	Pitch Feature Set	MLP			98.76
	Pitch Feature Set	SVM			99.38
	Pitch Feature Set	LR			<u>99.69</u>
Free_CNN	Raw speech signal	CNN + DNN	Yes	No	98.75
Sinc	Raw speech signal	CNN + DNN	Yes	Yes	99.38
Sinc2	Raw speech signal	CNN + DNN	Yes	Yes	98.75
Gamma	Raw speech signal	CNN + DNN	Yes	Yes	**100.0**
Gauss	Raw speech signal	CNN + DNN	Yes	Yes	98.13
Cascade	Raw speech signal	CNN + DNN	Yes	Yes	99.38

Table 6
Comparison between our proposed models with the literature works on the complete *TIMIT* test set.

Method	Input	Algorithm	end-to-end	Interpretable	Accuracy (%)
[12]	Traditional features	SVM	No	No	96.45
[26]	Traditional features	ensemble learning	No	No	98.42
[21]	Traditional features	LSTM	Yes	No	98.40
[28]	x-vector, d-vector	DNN	No	No	**99.60**
[35]	LCT	CNN + DNN	No	No	95.61
Free_CNN	Raw speech signal	CNN + DNN	Yes	No	<u>99.41</u>
Sinc	Raw speech signal	CNN + DNN	Yes	Yes	99.33
Sinc2	Raw speech signal	CNN + DNN	Yes	Yes	99.03
Gamma	Raw speech signal	CNN + DNN	Yes	Yes	<u>99.41</u>
Gauss	Raw speech signal	CNN + DNN	Yes	Yes	99.03
Cascade	Raw speech signal	CNN + DNN	Yes	Yes	<u>99.41</u>

interpretability aspects, as well as the accuracy of the different methods in experiments conducted on the complete TIMIT test set. For our own models, the reported accuracies represent the best result obtained from three runs of experiments. The best and second-best accuracies are highlighted in bold and underlined format, respectively.

Based on the results of Table 6, it is evident that our models achieve a comparable or superior accuracy of 99.41% compared to existing methods, with only one exception showing a difference of 0.19%. Notably, our models provide interpretability, which often results in reduced accuracy. Furthermore, it is worth highlighting that the best-performing method in terms of accuracy requires a two-step training process, one for the embedder and another for the classifier. In contrast, our method is designed to be end-to-end, allowing the training process to be performed in one step, simplifying the overall workflow. Regarding the types of filters, while *Free-CNN*, *Gamma,* and *Cascade* models show the best accuracy, it can be seen that other types of filters also show high efficiency compared to other approaches in the literature. These findings emphasize the effectiveness and versatility of our proposed models, which achieve high accuracy while maintaining interpretability and a simple training process. The performance of different filter types, including the consistently strong performance of *Cascade* filters, further highlights the robustness of our approach.

4.2.2. Learned representation

In this subsection, we have used t-SNE (t-distributed Stochastic Neighbor Embedding) analysis as a visualization method to further investigate the obtained results. The t-SNE projection method is a well-known unsupervised non-linear dimensionality reduction technique that can be used for visualizing high-dimensional data [47]. It maps high-dimensional points into a low-dimensional space based on a stochastic selection of similar neighbors in such a way that similar samples are modeled with nearby points, and dissimilar ones are far from each other with high probability. In this study, t-SNE is used to evaluate the representation power of the examined models for test samples. The result of applying t-SNE on the representation learned from the third convolutional layer is shown in Fig. 4. Since this is an unsupervised approach, sample labels are ignored during the dimensionality reduction process, and they are colored after applying the method. Furthermore, we also showcase the utilization of MFCC features as a robust representation of traditional features. The result indicates that the representation learned by the CNN models creates more distinct compact regions for male and female classes. Here, the *Cascade* model has the best representation in terms of class compactness and linear separability.

4.2.3. Accumulative magnitude response

Physiological differences between males and females can lead to variations in their acoustical parameters. Females generally have a higher fundamental frequency (pitch) range than males. Additionally, it has been observed that the average formant frequencies of females are approximately 20% higher than those of males. This difference is attributed to the inverse relationship between the overall length of the vocal tract and the formant frequency. As a result, a crucial feature that distinguishes males from females is the formant frequencies in vowels they produce, with females exhibiting a higher range of frequencies [48–51].

To evaluate the frequency sensitivity of the learned filter bank using different AFMs, the cumulative magnitude response of all filters learned in the first convolutional layer is depicted in Fig. 5. It is known that the first three formants are important for differentiating between male and female speech, and they are characterized by one or two consecutive peaks in almost all models. Interestingly, while the cumulative magnitude response of the *Free-CNN* model is noisy, the other models, especially the AFMs such as *Gaussian* (Gauss) and *gammaton* (Gamma), concentrate on specific frequencies, indicating that the use of such models makes the network more sensitive to important frequency regions and leads to better performance.

4.2.4. Histogram of center frequencies

Mel-filter bank is a result of analyzing the properties of the human auditory system. In this design, the filters that operate at low frequencies are more numerous than those that operate at high frequencies. Moreover, the filters that operate in lower frequency regions have sharper peaks, while those in high regions are shallower. To compare the distribution of the number of learned filters in different frequency regions,

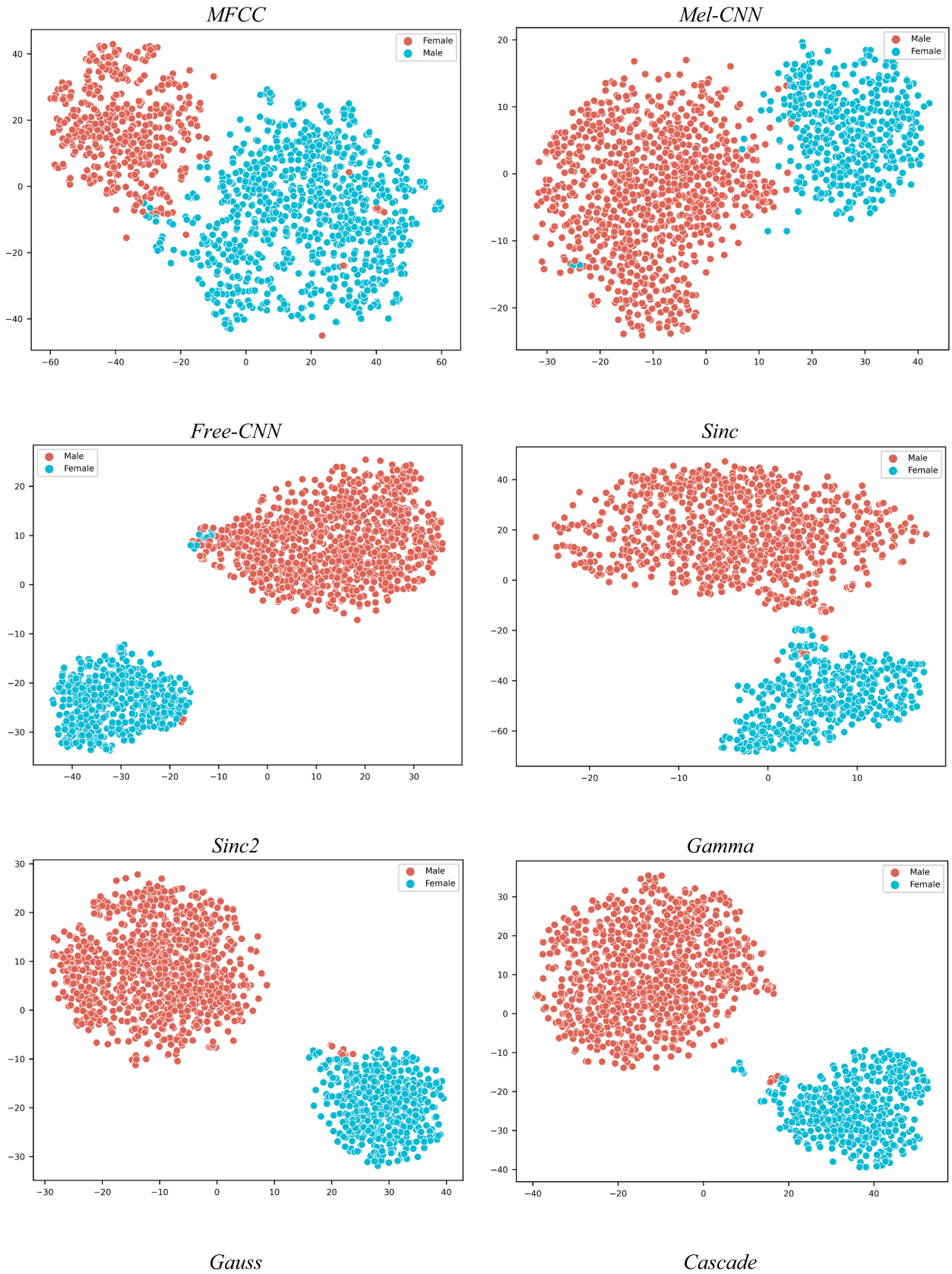


Fig. 4. T-SNE projection of extracted features from convolutional layers for test samples.

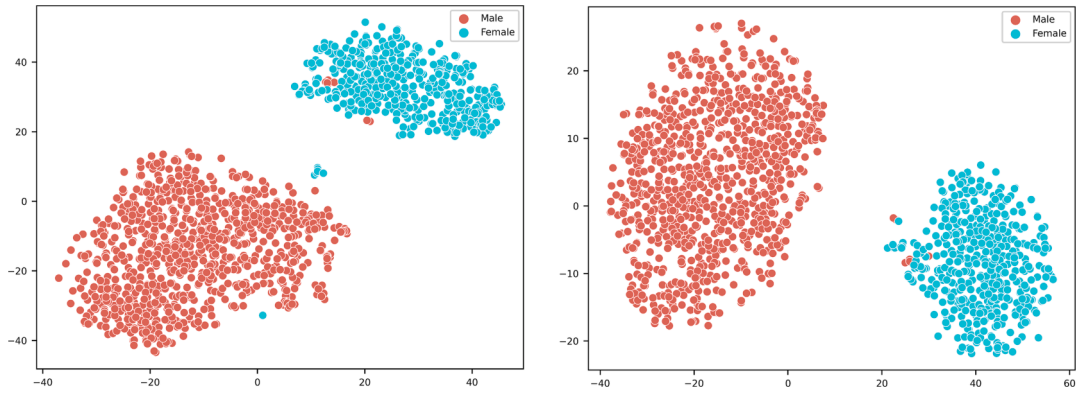


Fig. 4. (*continued*).

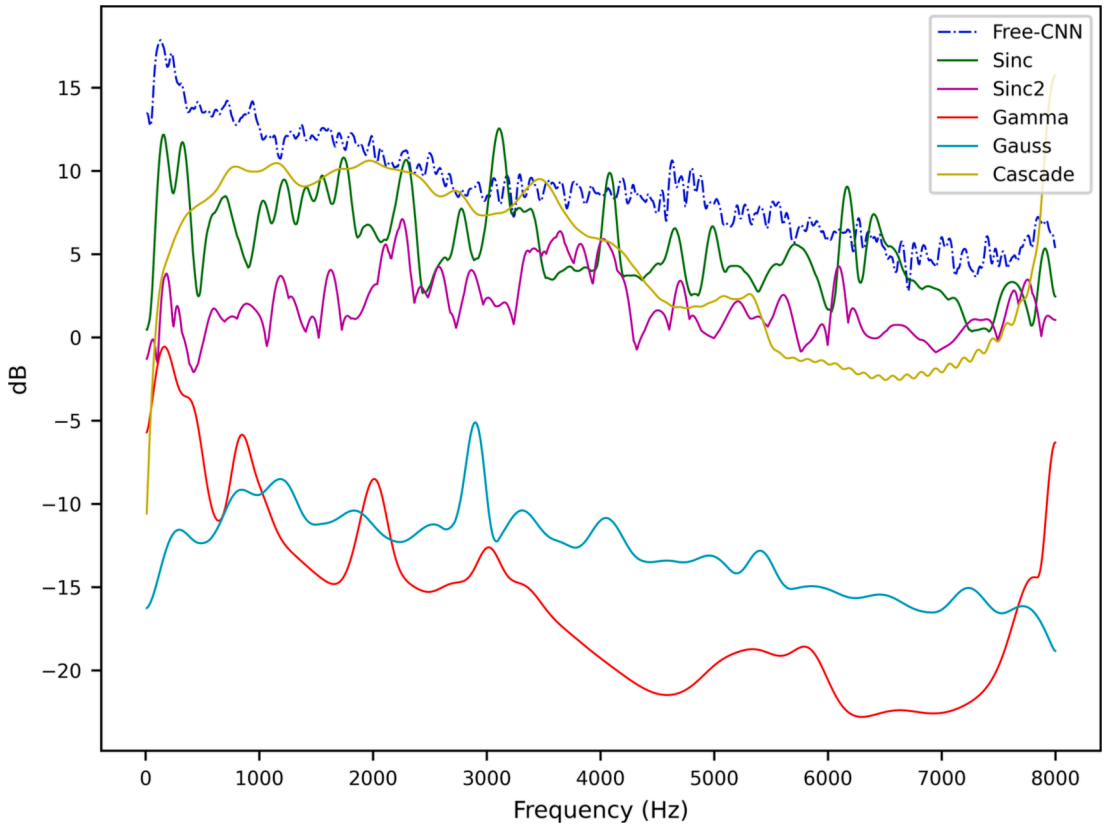


Fig. 5. Accumulative magnitude response of learned filter bank for all models.

the histogram of the center frequency of the learned filters in different models is compared with the *Mel*-scale filter bank in Fig. 6. While the overall trend of learned filter banks is similar to the *Mel*-scale, there are slight differences in some frequency regions. For example, the number of filters that operate in the 3, 4, and 5 kHz regions is greater than *Mel*-scale filter bank. This is due to the extra impact of the formants in determining the gender of the speaker.

4.2.5. *Quality factor*

The Quality Factor (QF), defined as the ratio of the center frequency to the bandwidth of a filter, is a good measure for evaluating both parameters at the same time. Fig. 7 displays the QF of the filters learned in all models. To have an easy comparison in one picture, the QF of the networks using AFMs is projected onto the fitted linear curve. As observed in previous figures, more filters operate in low frequencies while fewer operate in high frequencies. In other words, higher frequencies are not as important in recognizing the gender of a speaker from the speech signals. All filter types, in some way, disregard the frequency region around 4.5, 5.5, and 7 kHz, which suggests that these frequency regions do not provide sufficient information about the speaker's gender.

4.3. *Examining the impact of mean-shift clustering*

In Section 3, the mean-shift clustering technique was introduced to reduce the number of filters in the first convolutional layer. Here, the filters learned by the *Cascade* model are mapped into the complex plane using the method explained earlier. The reason for choosing the *Cascade* model was that it obtained better results than other ones. Since the cascade filter results from cascading two Gaussian filters, each filter is represented by two poles. This means that the 80 cascade filters give 160 pole locations in the complex plane. The bandwidth parameter of the

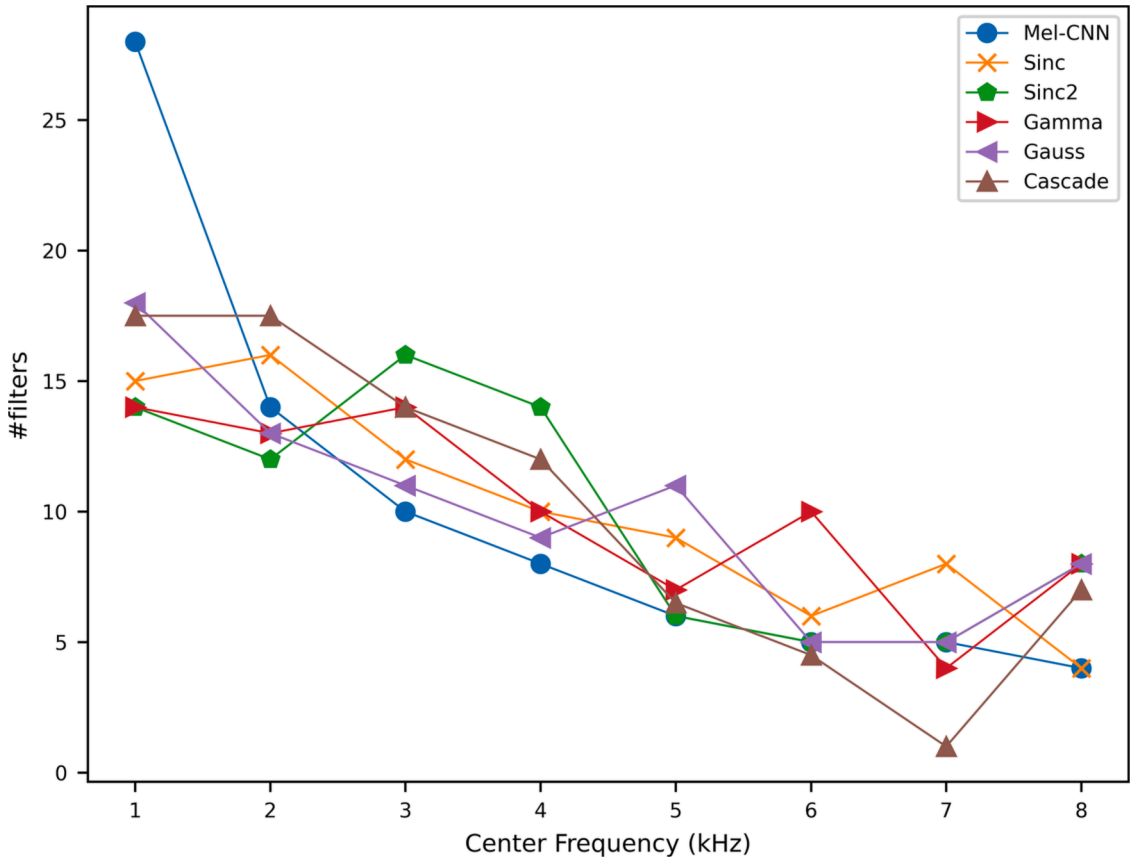


Fig. 6. Histogram of center frequencies for all models.

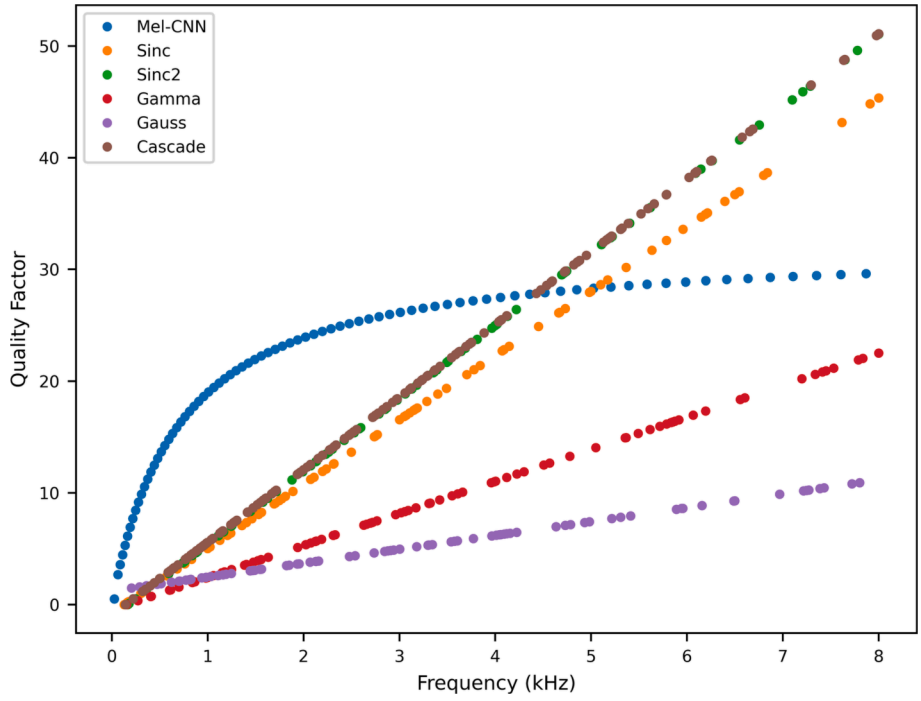


Fig. 7. Quality factor of the filter bank learned by different models.

mean-shift clustering algorithm is set in such a way that 40 clusters be created for these 160 pole locations. The resulting 40 cluster centers are used to create a new triangular filter bank for the first convolutional layer. A *triangular* filter bank is here used to have a fair comparison with the *Mel-CNN* model. The CNN model is trained again with this new filter set while the convolutional layers are fixed during the training process. Using this new filter bank, which is named *Reduced* hereafter, led to a CER of 0.97 $\pm$ 0.08, which is the second-best result after the *Cascade*

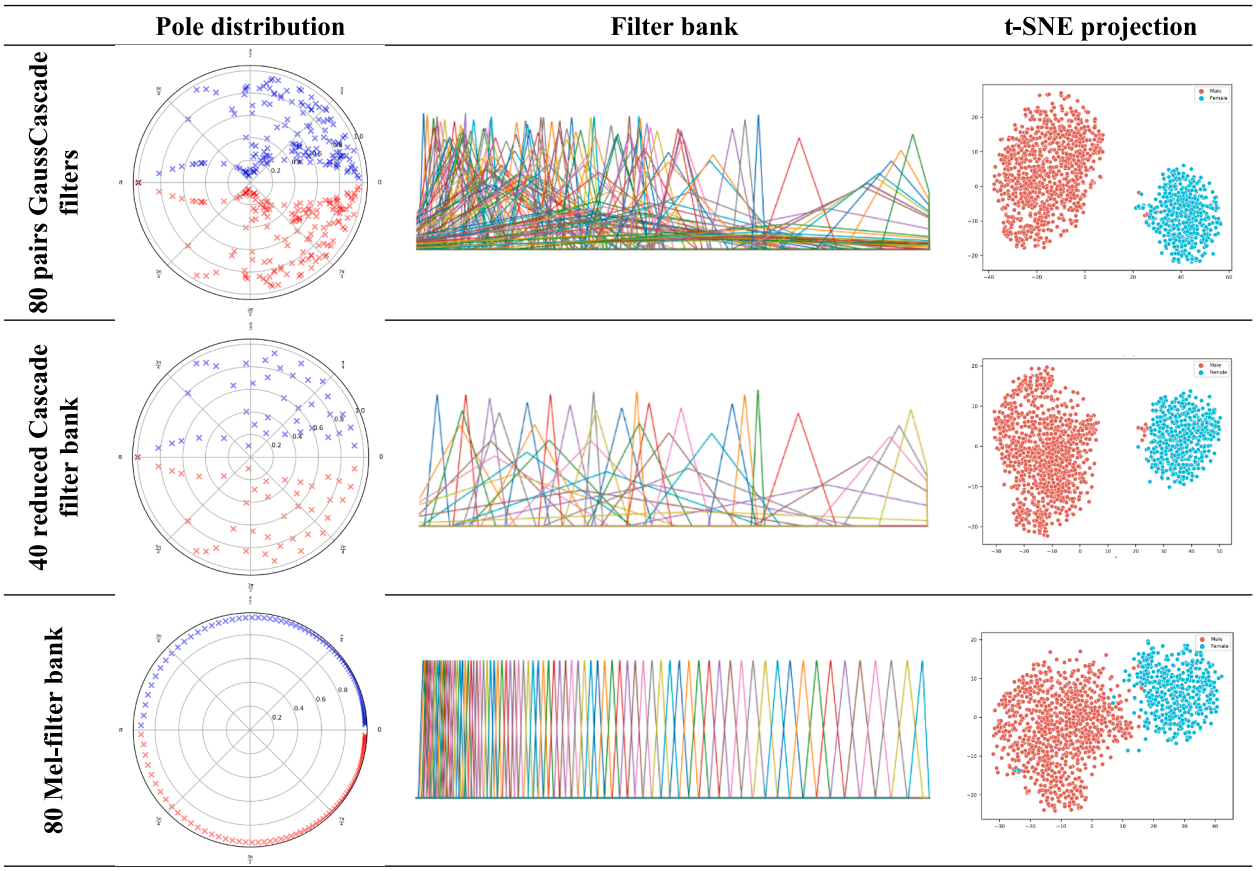


Fig. 8. Comparison of the filter bank and respective pole distribution in the complex plane and the t-SNE projection of learned representation for three situations, 1) *Cascade* model, 2) *Reduced* model, and 3) *Mel-CNN* model.

model, and an FER of 3.94 ± 0.43, which is the best result after the *Free-CNN* model.

Fig. 8 displays the pole distribution of the learned filters in the *Cascade* model, as well as the filters used in the *Mel-CNN* and the reduced model. It also shows their filter banks and t-SNE projections of the learned representation of each model. The representation learned by the reduced model is as good as the *Cascade* model, while it utilizes fewer parameters.

4.4. Examining the generalizability of the models

In order to assess the effectiveness of the models in real-world scenarios with varying noise conditions, several experiments are conducted here. It should be noted that the *TIMIT* dataset used for training the models in this study primarily consists of studio-quality recordings. As a result, the models trained on this dataset exhibit poorer performance when applied to unconstrained environments. This highlights the need to evaluate and refine the models in diverse and unconstrained settings to ensure their robustness and generalizability.

While the speech signals in *LibriSpeech* are generally considered clean, it is important to note that the recording devices and channel conditions can vary significantly among different utterances and speakers in the dataset. Furthermore, since *LibriSpeech* is composed of read English speech, the tone, and style of each utterance are different for each speaker. In light of these factors, we utilize the *LibriSpeech* dataset to fine-tune the best model of the previous three runs for another 200 epochs. Subsequently, we evaluate the resulting models on previously unseen data from the *VoxCeleb1* and *RAVDESS* datasets to evaluate their performance in more realistic and diverse scenarios.

Table 7 compares the literature methods evaluated in [11] with a subset of *VoxCeleb* and our models. Since the accuracies reported in [11]

Table 7
Comparison between our proposed models with the literature works on the *VoxCeleb* dataset.

Method	Input	Algorithm	End-to-End	Interpretable	Accuracy (%)
[20]	Raw speech signal	CNN + DNN	Yes	No	~ 61.8
[25]	log-*Mel* filter bank coefficients	CNN + DNN	Yes	No	~ 87.3
[21]	Traditional features	LSTM	Yes	No	~ 55.3
[27]	Raw speech signal	CNN + LSTM + DNN	No	No	~ 78.9
[11]	Spectrogram	CNN + LSTM + DNN	Yes	No	96.23
Free_CNN	Raw speech signal	CNN + DNN	Yes	No	97.48
Sinc	Raw speech signal	CNN + DNN	Yes	Yes	**97.66**
Sinc2	Raw speech signal	CNN + DNN	Yes	Yes	96.33
Gamma	Raw speech signal	CNN + DNN	Yes	Yes	97.17
Gauss	Raw speech signal	CNN + DNN	Yes	Yes	96.74
Cascade	Raw speech signal	CNN + DNN	Yes	Yes	97.27
Reduced	Raw speech signal	CNN + DNN	Yes	Yes	97.44

are presented in bar chart diagram format, we estimate the values based on the figures provided. The best-performing method is highlighted in bold, while the second-best method is underlined. Although the *VoxCeleb1* dataset contains a variety of ethnicities, accents, professions, and ages from real word utterances, all versions of our models consistently outperform the other methods. This highlights the robustness and effectiveness of our models in handling real-world utterances from varied speakers and scenarios.

Based on the results of Table 7, when comparing the performance of different filter types, it becomes evident that the *Sinc* and *Free_CNN* models outperform the others. Notably, the high accuracy achieved by the *Reduced* model demonstrates that reducing the number of filters can result in a more generalized model while simultaneously reducing the number of model parameters.

We also compare the performance of the proposed models with state-of-the-art models in gender recognition using the RAVDESS dataset. The results of this comparison are reported in Table 8. It is worth noting that our models were exclusively trained on the *TIMIT* and *Librispeech* datasets. Despite this limitation, our models have achieved remarkable results, surpassing the performance of all other approaches considered in this study. This outcome clearly demonstrates the impact of incorporating AFMs in front of the CNN, which enhances the generalization capabilities of deep models.

Based on the results of Table 8, the *Cascade* model stands out as the top-performing auditory filter model, surpassing all others, while the *Gauss* model ranks as the least effective. Interestingly, the *Reduced* model achieves competitive accuracy compared to the *Sinc* and even outperforms the *Gauss*, *Sinc2*, and *Free_CNN* models. This notable outcome underscores the effectiveness of the filter pruning process and emphasizes the importance of adopting similar approaches to reduce the number of filters. By using filter pruning techniques, we can increase the efficiency of the model while maintaining competitive performance in various tasks.

Table 8
Comparison between our proposed models with the literature works on the *RAVDESS* dataset.

Method	Input	Algorithm	End-to-End	Interpretable	Accuracy (%)
[20]	Raw speech signal	CNN + DNN	Yes	No	84.35
[29]	Spectrogram	CNN + DNN	Yes	No	86.85
[21]	Traditional features	LSTM	Yes	No	86.79
[17]	Traditional features	SVM	No	No	90.61
[23]	Spectrogram	ResNet	Yes	No	91.50
[15]	Traditional features	SVM, KNN	No	No	92.50
[34]	Acoustic features	CNN + GRU	No	No	87.13
Free_CNN	Raw speech signal	CNN + DNN	Yes	No	93.54
Sinc	Raw speech signal	CNN + DNN	Yes	Yes	96.32
Sinc2	Raw speech signal	CNN + DNN	Yes	Yes	94.86
Gamma	Raw speech signal	CNN + DNN	Yes	Yes	<u>97.78</u>
Gauss	Raw speech signal	CNN + DNN	Yes	Yes	93.19
Cascade	Raw speech signal	CNN + DNN	Yes	Yes	**97.99**
Reduced	Raw speech signal	CNN + DNN	Yes	Yes	96.11

5. Conclusions and future works

This paper proposed a novel interpretable architecture to improve the performance of a typical CNN architecture for the SGR task by utilizing AFMs in the first convolutional layer. This technique not only reduced the number of model parameters but also resulted in better performance compared to the traditional convolutional layers. By applying constraints to the first convolutional layer, the model is compelled to acquire a more generalized representation of the input data. Consequently, incorporating AFMs further enhances the model's capacity for generalization, interpretability, and the acquisition of improved representations within the convolutional layers. This process contributes to the overall effectiveness and robustness of the model, allowing it to learn and extract meaningful features from the data, ultimately leading to enhanced performance in the SGR task.

By mapping the learned filters into the complex plane and clustering them, a reduced filter bank was obtained, which can be used as the filter bank of the first convolutional layer for a new model. The resulting new model, which was named the *Reduced* model, showed a good performance in terms of the number of model parameters, overall accuracy, and rich learned representation. Furthermore, the success of the *Reduced* model in achieving high accuracy on the *VoxCeleb1* and RAVDESS datasets shows that by reducing the complexity of the model, we can increase its generalization capabilities, enabling it to perform well on previously unseen data. This highlights the trade-off between model complexity and generalization, indicating that a streamlined model with fewer parameters can still achieve competitive levels of accuracy. In other words, by simplifying the architecture and reducing the number of parameters, we enable the model to capture essential patterns and features in a more generalized manner. The lightweight nature and generalizability of the model are crucial for real-world applications, particularly when considering its potential deployment on low-resource devices.

Future work will focus on learning a fixed general filter bank that is suitable for use in MFCC for gender recognition rather than relying on a *Mel*-scaled filter bank. In this approach, the impact of other filters learned in the second and third convolutional layers will need to be considered.

CRediT authorship contribution statement

Hossein Fayyazi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **Yasser Shekofteh:** Conceptualization, Investigation, Validation, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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